Question 7		
(a)	Below the threshold frequency, no electrons are emitted (no matter the intensity). Light interacts with the electrons as a photon (particle) of energy h*f*. h*f* must be greater than the metal's work function to liberate an electron.	**B1** **B1**
(b)	There is significant diffraction when the wavelength is comparable to or larger than the object's dimensions. Visible light has wavelength of 380 nm to 750 nm, while an electron (beam's kinetic energy could be adjusted to give it a) wavelength smaller than 140 nm.	**B1** **B1**
(c)	According to de Broglie's relation, photons have momentum $p = \frac{h}{\lambda}$. They experience a change in momentum when they bounce off the sail. The force that the sail exerts on them is equal to the force they exert on the sail.	**B1** **B1**